

Jacob Neumann

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Employment

- 2025 – present  **Postdoctoral Researcher**, Reykjavik University, School of Computer Science
Supervisor: Tarmo Uustalu
- 2021 – 2024  **Tutor** (teaching assistant), University of Nottingham, School of Computer Science
Modules supported:
- *Introduction to Formal Reasoning* (lecturer: Thorsten Altenkirch): Second-year undergraduate module covering the basics of interactive theorem proving in Lean
 - *Programming Paradigms* (lecturer: Graham Hutton): First-year undergraduate module on elementary functional programming in Haskell
 - *Languages and Computation* (lecturer: Thorsten Altenkirch): Second-year undergraduate module on automata and computability theory, with optional content on Lean formalization
- 2020, 2021  **Visiting Lecturer**, Carnegie Mellon University, School of Computer Science
Course: *Principles of Functional Programming*
Co-instructor (2020): Dilsun Kaynar
Delivered 21 lectures (2020) and 32 lectures (2021), covering introductory functional programming in Standard ML; oversaw delivery and grading of homework assignments; interviewed, selected, and managed course staffs of 23 teaching assistants (2020) and 15 teaching assistants (2021)
Number of students: approx. 110 (2020), approx. 80 (2021)

Education

- 2021 – 2025  **Ph.D. Computer Science**
University of Nottingham, School of Computer Science, Functional Programming Lab
Thesis title: *A Generalized Algebraic Theory of Directed Equality*
Supervisor: Thorsten Altenkirch
- 2019 – 2020  **M.Sc. Logic, Computation and Methodology**
Carnegie Mellon University, Department of Philosophy
Thesis title: *Semantics of Nondeterministic Construction*
Supervisor: Adam Bjorndahl
- 2015 – 2019  **B.Sc. Discrete Mathematics and Logic**
Carnegie Mellon University, Department of Mathematics
Additional Major: Logic and Computation (Dept. of Philosophy)
Mellon College of Science Honors

Publications

- 2025  **Synthetic 1-Categories in Directed Type Theory**
in Rasmus Ejlers Møgelberg and Benno van den Berg editors, proceedings of the 30th International Conference on Types for Proofs and Programs (TYPES 2024)
Leibniz International Proceedings in Informatics (LIPIcs), Schloss Dagstuhl – Leibniz-Zentrum für Informatik, Dagstuhl, Germany

Grants and Awards

- June 2024 **Second-Order Generalised Algebraic Theories for Modal and Substructural Type Theories**
Short-Term Scientific Mission
Funding organization: EU COST Action CA20111 (EuroProofNet)
Host: Ambrus Kaposi, Eötvös Loránd University, Budapest, Hungary
- May 2025 **Dinatural Semantics of Directed Type Theory**
Short-Term Scientific Mission
Funding organization: EU COST Action CA20111 (EuroProofNet)
Host: Niccolò Veltri, Tallinn University of Technology, Tallinn, Estonia